



Experiment

AWS- Cloud Solutions Architect

Slot: A14+D11+D12

Winter Semester: 2025-2026

Virtual Private Cloud

SUBMITTED BY:

Dhruv Panwar

23BEY10057

SUBMITTED TO:

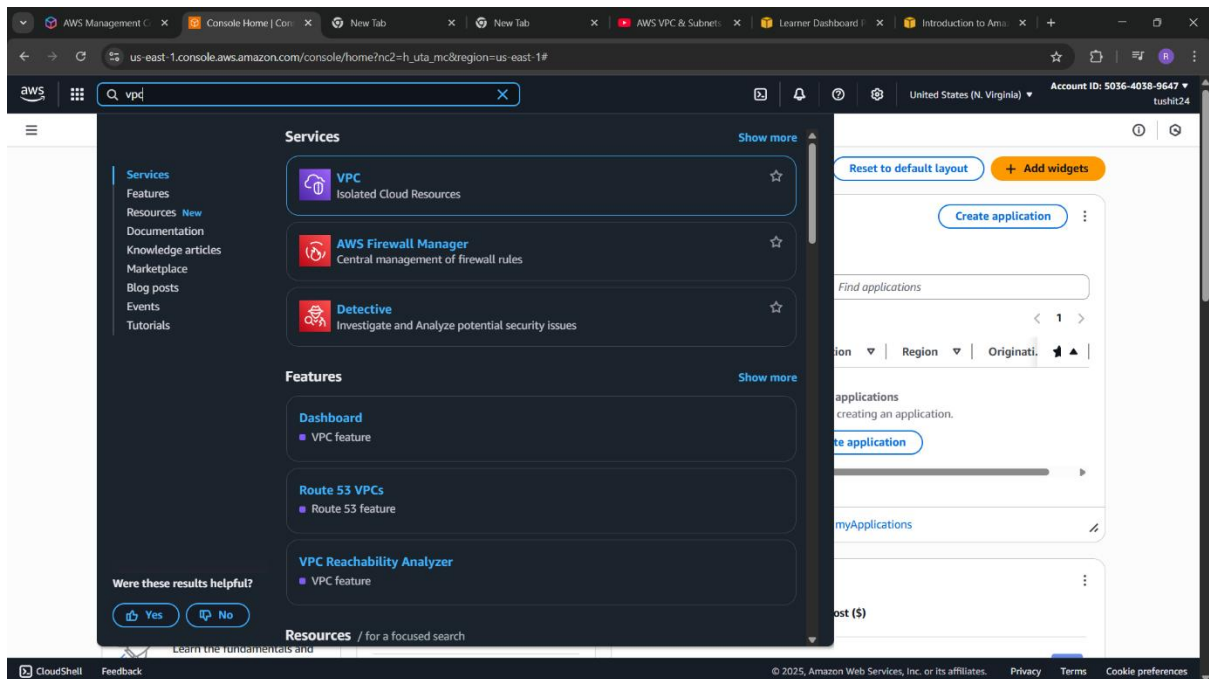
Dr. Anju Shukla

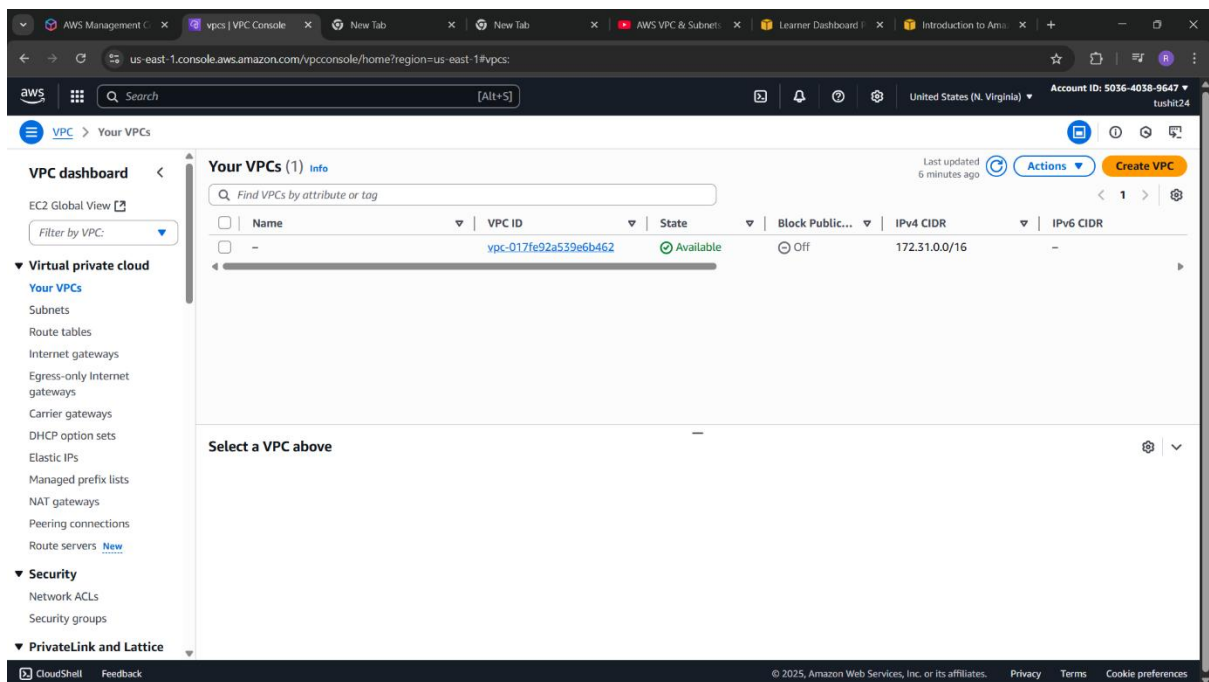
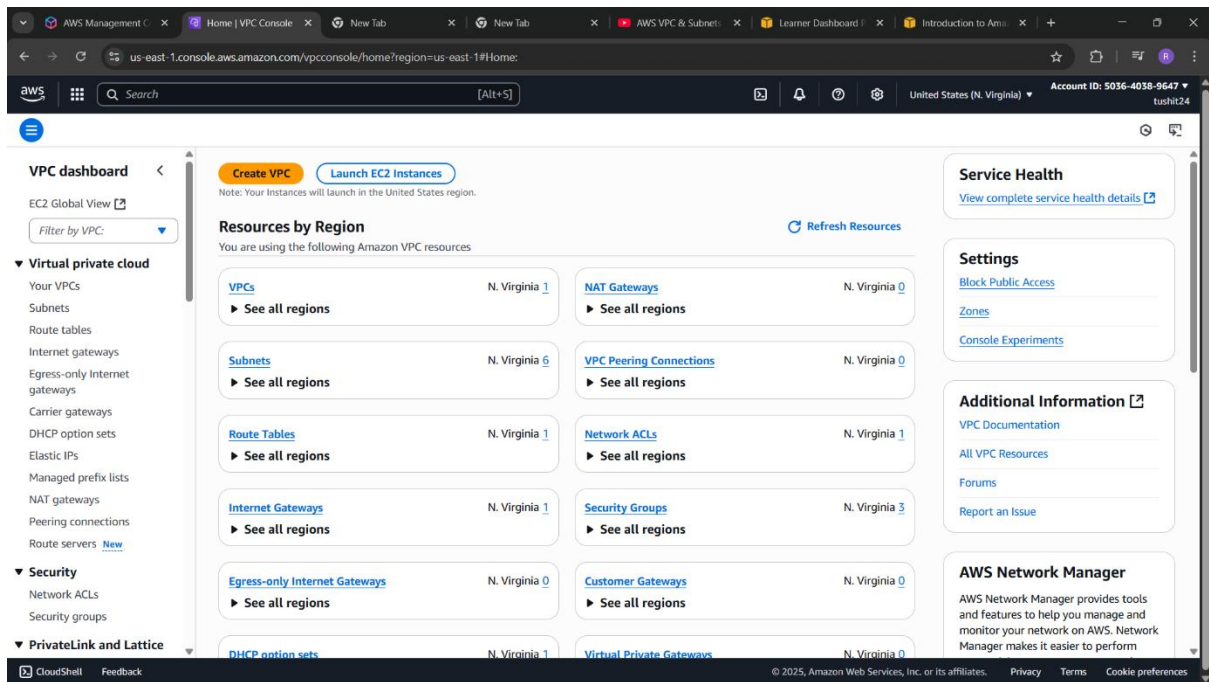
Experiment

AIM: Create a Virtual Private Cloud in AWS and create a EC2 in the same VPC

VPC Creation

1. Log in to AWS Management Console.
2. Navigate to **VPC Dashboard**.
3. Click on **Create VPC**.
4. Select **VPC only** option.
5. Enter VPC name and IPv4 CIDR block (e.g., 10.0.0.0/16).

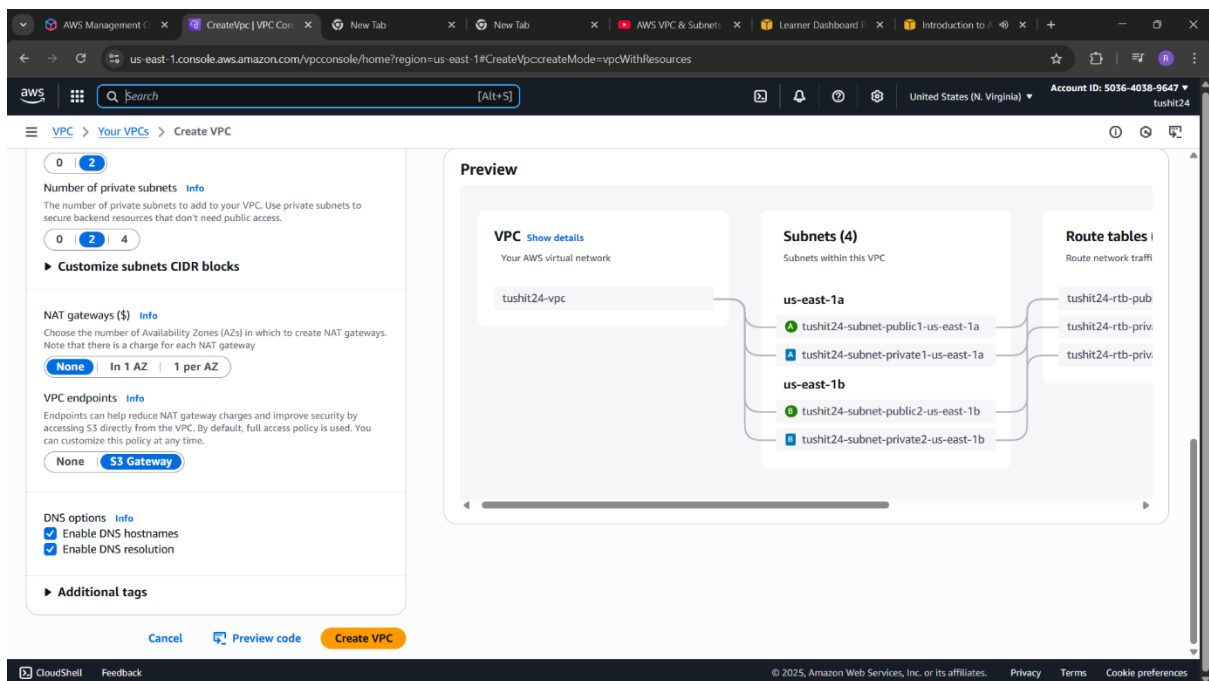
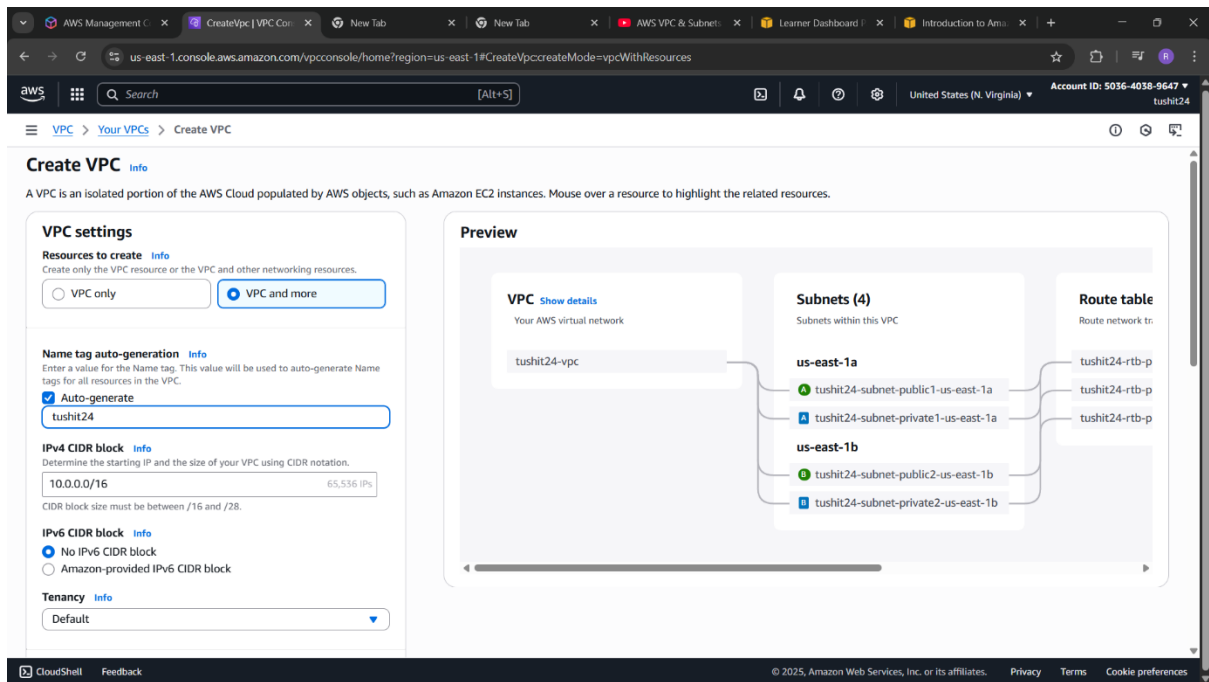




Configure VPC Settings

Steps:

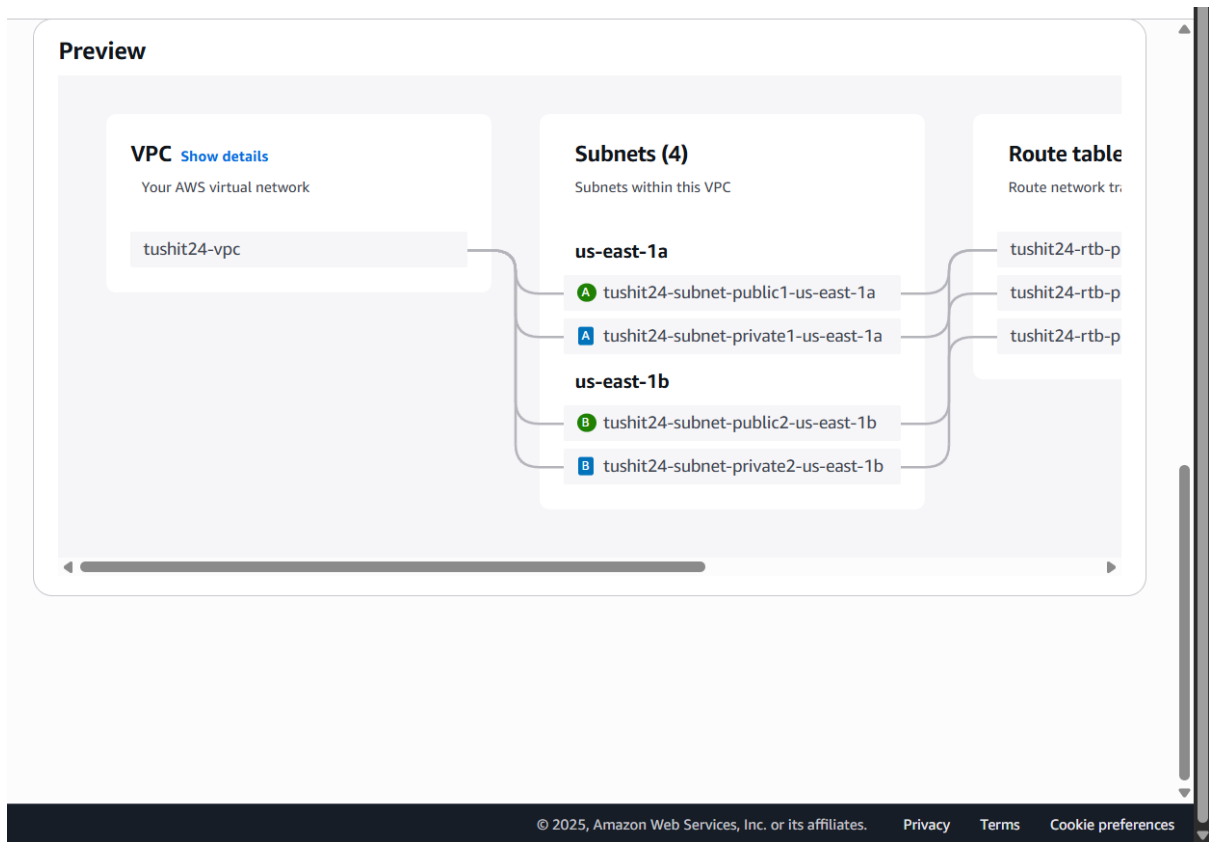
1. Keep default tenancy as **Default**.
2. Enable DNS hostname if required.
3. Review all configurations.
4. Click on **Create VPC** button.
5. Wait until VPC is successfully created.



Create Subnet in VPC

Steps:

1. Go to **Subnets** in VPC dashboard.
2. Click on **Create Subnet**.
3. Select the created VPC.
4. Enter subnet name and CIDR block (e.g., 10.0.1.0/24).
5. Click on **Create Subnet**.



AWS Management | VPC | us-east-1 | New Tab | New Tab | AWS VPC & Subnet | Learner Dashboard | Introduction to Am... | + | - | [AWS Logo] | Search | [Alt+S] | United States (N. Virginia) | Account ID: 5036-4038-9647 | tushit24

VPC > Your VPCs > Create VPC > Create VPC resources

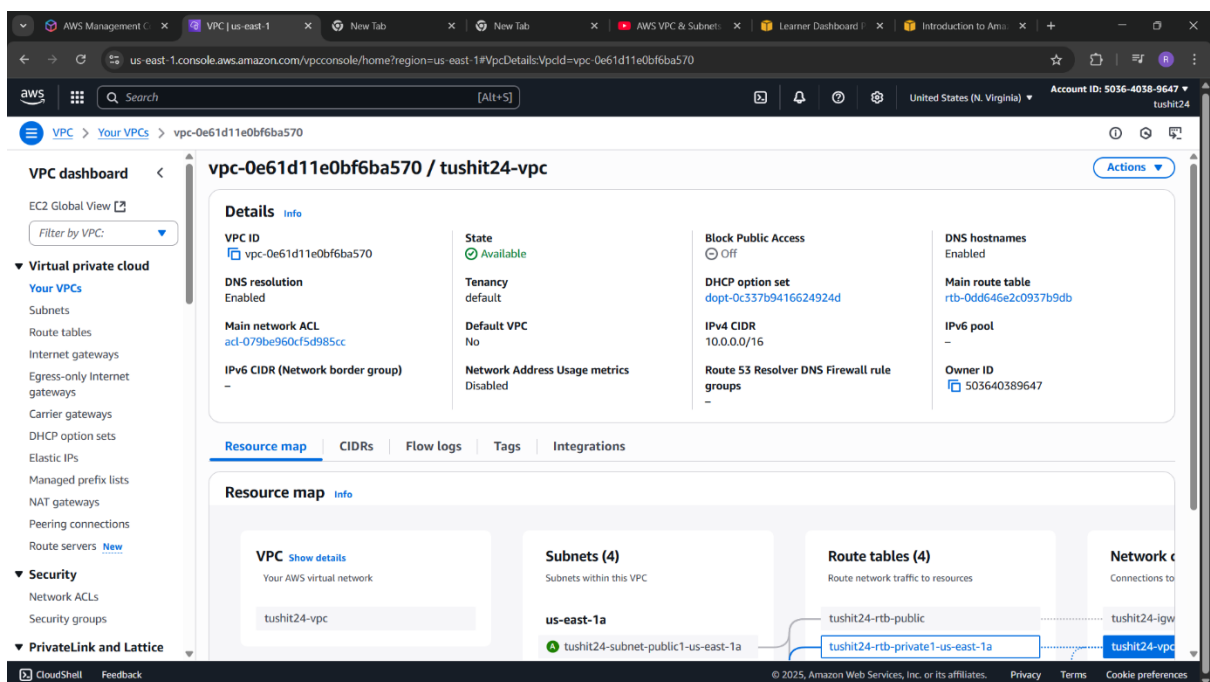
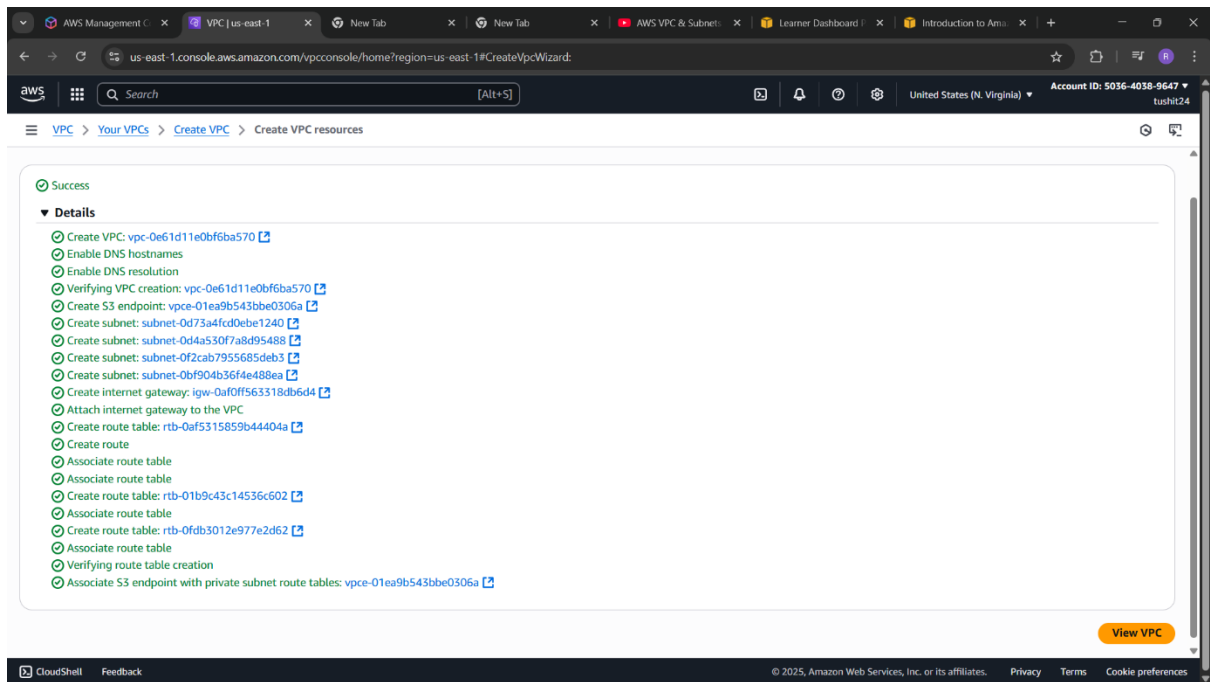
Create VPC workflow

Verifying VPC creation: vpc-0e61d11e0bf6ba570 14%

▼ Details

- ✔ Create VPC: vpc-0e61d11e0bf6ba570
- ✔ Enable DNS hostnames
- ✔ Enable DNS resolution
- Verifying VPC creation: vpc-0e61d11e0bf6ba570
- Create S3 endpoint
- Create subnet
- Create subnet
- Create subnet
- Create subnet
- Create internet gateway
- Attach internet gateway to the VPC
- Create route table
- Create route
- Associate route table
- Associate route table
- Create route table
- Associate route table
- Create route table
- Associate route table
- Verifying route table creation
- Associate S3 endpoint with private subnet route tables

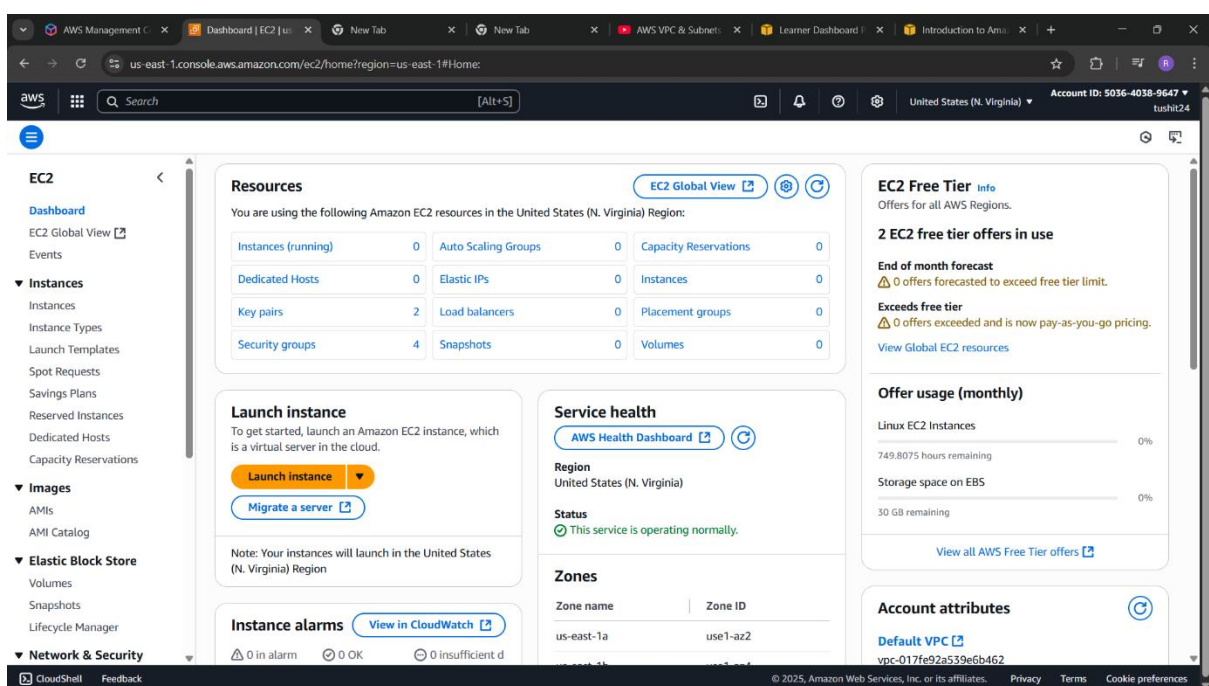
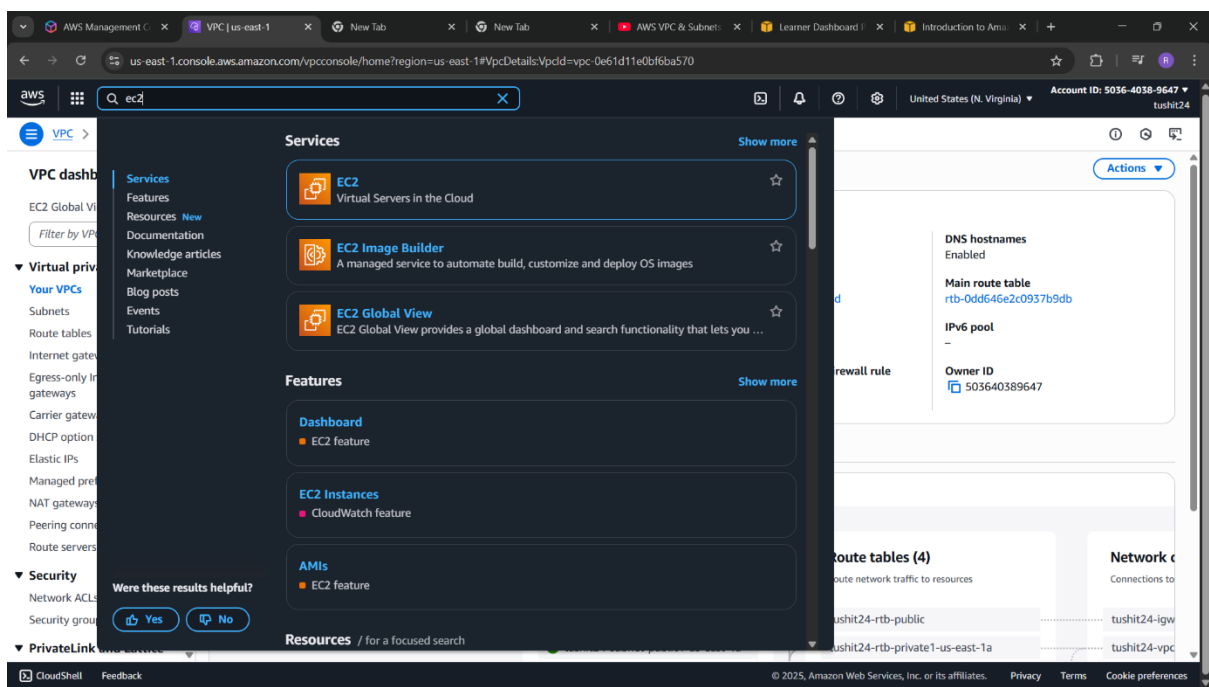
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Launch EC2 Instance

Steps:

1. Navigate to **EC2 Dashboard**.
2. Click on **Launch Instance**.
3. Choose an Amazon Machine Image (AMI).
4. Select instance type (e.g., t2.micro).
5. Under network settings, select the created VPC and subnet.



us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#LaunchInstances:

Search

United States (N. Virginia)Account ID: 5036-4038-9647tushit24

EC2 > Instances > Launch an instance

It seems like you may be new to launching instances in EC2. Take a walkthrough to learn about EC2, how to launch instances and about best practices.

Take a walkthroughDo not show me this message again.

Launch an instance

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags

Name

tushit24

Add additional tags

Application and OS Images (Amazon Machine Image)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Search our full catalog including 1000s of application and OS images

Recents

Quick Start

Amazon Linux

macOS

Ubuntu

Windows

Red Hat

SUSE Linux

Debian

Browse more AMIs

Including AMIs from AWS, Marketplace and

Summary

Number of instances 1

Software Image (AMI)
Amazon Linux 2023 AMI 2023.8.2...[read more](#)
ami-0de716d6197524dd9

Virtual server type (instance type)
t2.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

Free tier:

In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4

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us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#LaunchInstances:

Search

United States (N. Virginia)Account ID: 5036-4038-9647tushit24

EC2 > Instances > Launch an instance

Instance type

Instance type

t2.micro

Family: t2 1 vCPU 1 GiB Memory Current generation: true

On-Demand Windows base pricing: 0.0162 USD per Hour

On-Demand Ubuntu Pro base pricing: 0.0134 USD per Hour

On-Demand SUSE base pricing: 0.0116 USD per Hour

On-Demand RHEL base pricing: 0.026 USD per Hour

On-Demand Linux base pricing: 0.0116 USD per Hour

Free tier eligible

All generations

Compare instance types

Additional costs apply for AMIs with pre-installed software

Key pair (login)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - **required**

Proceed without a key pair (Not recommended)

Default value

Create new key pair

Network settings

Network

vpc-017fe92a539e6b462

Subnet

No preference (Default subnet in any availability zone)

Auto-assign public IP

Enable

Edit

Summary

Number of instances 1

Software Image (AMI)
Amazon Linux 2023 AMI 2023.8.2...[read more](#)
ami-0de716d6197524dd9

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Cancel

Launch instance

Preview code

CloudShellFeedback

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us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#LaunchInstances:

Search

United States (N. Virginia) Account ID: 5036-4038-9647 tushit24

EC2 > Instances > Launch an instance

Key pair (required)

Proceed without a key pair (Not recommended) Default value Create new key pair

Network settings

VPC - required

vpc-0e61d11e0bf6ba570 (tushit24-vpc) 10.0.0.0/16

Subnet

subnet-0f2cab7956685deb3 tushit24-subnet-private1-us-east-1a VPC: vpc-0e61d11e0bf6ba570 Owner: 503640389647 Availability Zone: us-east-1a Zone type: Availability Zone IP addresses available: 4091 CIDR: 10.0.128.0/20

Create new subnet

Auto-assign public IP

Disable

Firewall (security groups)

Create security group Select existing security group

Security group name - required

launch-wizard-2

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and _-./[!@.#%^&*(){}~`|;:'",<>?]{+}*

Summary

Number of instances

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.8.2...read more ami-0de716d6197524dd9

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t2.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

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Cancel Launch instance Preview code

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us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#LaunchInstances:

Search

United States (N. Virginia) Account ID: 5036-4038-9647 tushit24

EC2 > Instances > Launch an instance

Key pair (required)

Proceed without a key pair (Not recommended) Default value Create new key pair

Network settings

VPC - required

vpc-0e61d11e0bf6ba570 (tushit24-vpc) 10.0.0.0/16

Subnet

subnet-0d73a4fcd0be1240 tushit24-subnet-public1-us-east-1a VPC: vpc-0e61d11e0bf6ba570 Owner: 503640389647 Availability Zone: us-east-1a Zone type: Availability Zone IP addresses available: 4091 CIDR: 10.0.0.0/20

Create new subnet

Auto-assign public IP

Disable

Firewall (security groups)

Create security group Select existing security group

Security group name - required

launch-wizard-2

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and _-./[!@.#%^&*(){}~`|;:'",<>?]{+}*

Summary

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Amazon Linux 2023 AMI 2023.8.2...read more ami-0de716d6197524dd9

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t2.micro

Firewall (security group)

New security group

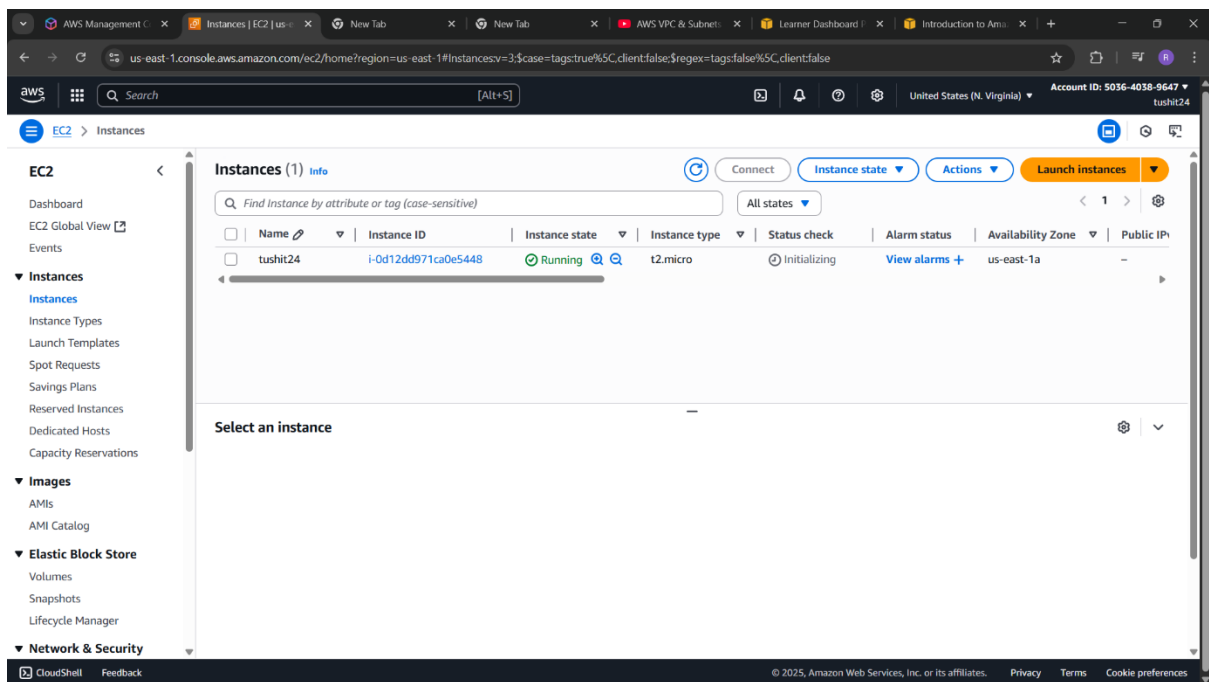
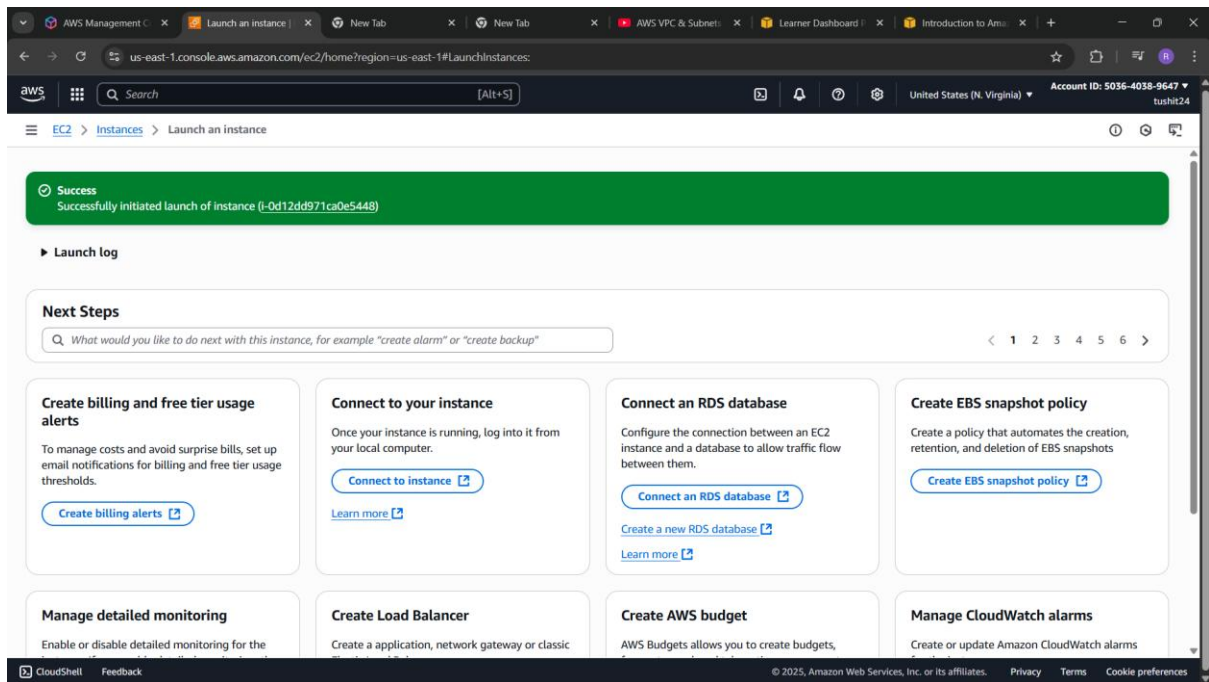
Storage (volumes)

1 volume(s) - 8 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4

Cancel Launch instance Preview code

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us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#Instances?v=3&case=tags:true%5C,client:false,\$regex=tags:false%5C,client:false

EC2 > Instances

Instances (1/1) Info

Find instance by attribute or tag (case-sensitive) All states

☒	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IP
☒	tushit24	i-0d12dd971ca0e5448	Running	t2.micro	Initializing	View alarms +	us-east-1a	-

i-0d12dd971ca0e5448 (tushit24)

Answer private resource DNS name	Instance type	Elastic IP addresses
-	t2.micro	-
Auto-assigned IP address	VPC ID	AWS Compute Optimizer finding
-	vpc-0e61d11e0bf6ba570 (tushit24-vpc)	Opt-in to AWS Compute Optimizer for recommendations. Learn more
IAM Role	Subnet ID	Auto Scaling Group name
-	subnet-0d73a4fcd0ebe1240 (tushit24-subnet-public1-us-east-1a)	-
IMDSv2	Instance ARN	Managed
Required	arn:aws:ec2:us-east-1:503640189647:instance/i-0d12dd971ca0e5448	false

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us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#LaunchInstances:

EC2 > Instances > Launch an instance

Launch an instance

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags

Name

tushit24-private

Add additional tags

Application and OS Images (Amazon Machine Image)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Search our full catalog including 1000s of application and OS images

Recents Quick Start

Amazon Linux

macOS

Ubuntu

Windows

Red Hat

SUSE Linux

Debian

[Browse more AMIs](#)
Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.1 AMI

Free tier eligible

Summary

Number of instances 1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.8.2...[read more](#)
ami-0de716d6197524d09

Virtual server type (instance type)

t2.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4

Cancel Launch instance Preview code

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Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

Proceed without a key pair (Not recommended) Default value [Create new key pair](#)

Network settings [Info](#)

VPC - required [Info](#)

vpc-0e61d11e0bf6ba570 (tushit24-vpc) 10.0.0.0/16 [Create new VPC](#)

Subnet [Info](#)

subnet-0f2cab7955685deb3 tushit24-subnet-private1-us-east-1a [Create new subnet](#)

VPC: vpc-0e61d11e0bf6ba570 Owner: 503640389647 Availability Zone: us-east-1a Zone type: Availability Zone IP addresses available: 4091 CIDR: 10.0.128.0/20

Auto-assign public IP [Info](#)

Disable

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group ☐ Select existing security group

Security group name - required

launch-wizard-3

Summary

Number of instances [Info](#)

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.8.2...[read more](#)
ami-0de716d6197524dd9

Virtual server type (instance type)

t2.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4

[Cancel](#) [Launch instance](#) [Preview code](#)

allow access from known IP addresses only.

[Add security group rule](#)

Advanced network configuration

Configure storage [Info](#) Advanced

1x 8 GiB gp3 Root volume, 3000 IOPS, Not encrypted

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4

[Add new volume](#)

[Click refresh to view backup information](#)

The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies.

0 x File systems [Edit](#)

Advanced details [Info](#)

Summary

Number of instances [Info](#)

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.8.2...[read more](#)
ami-0de716d6197524dd9

Virtual server type (instance type)

t2.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

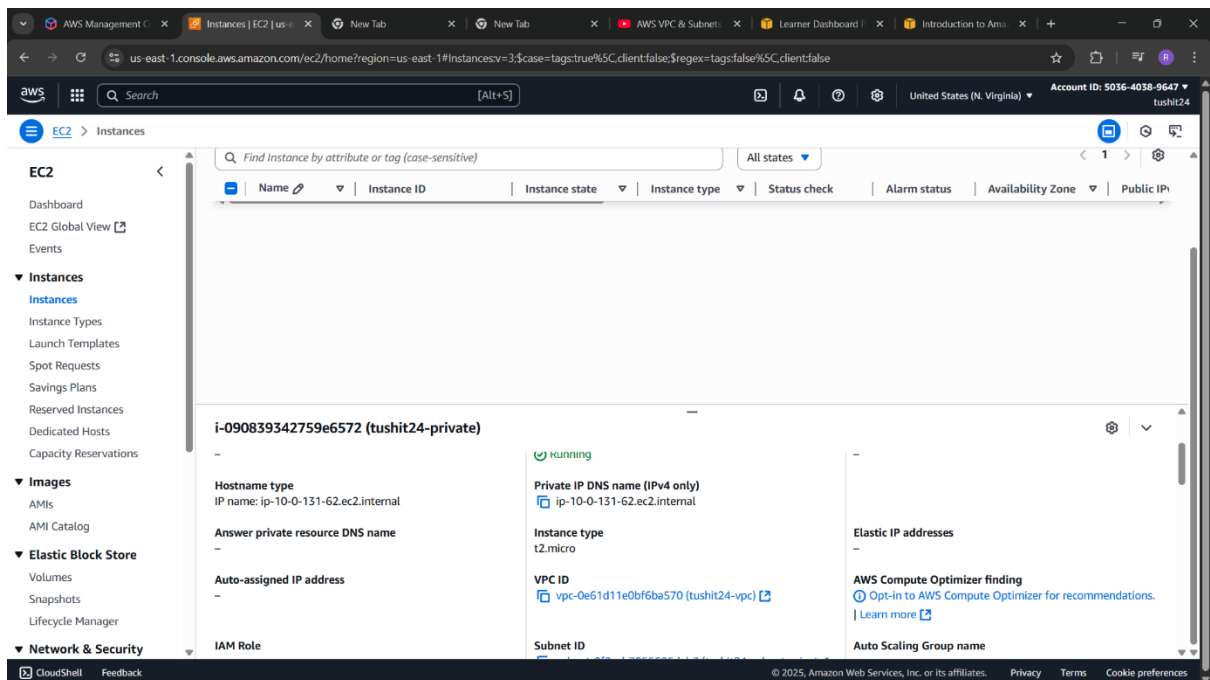
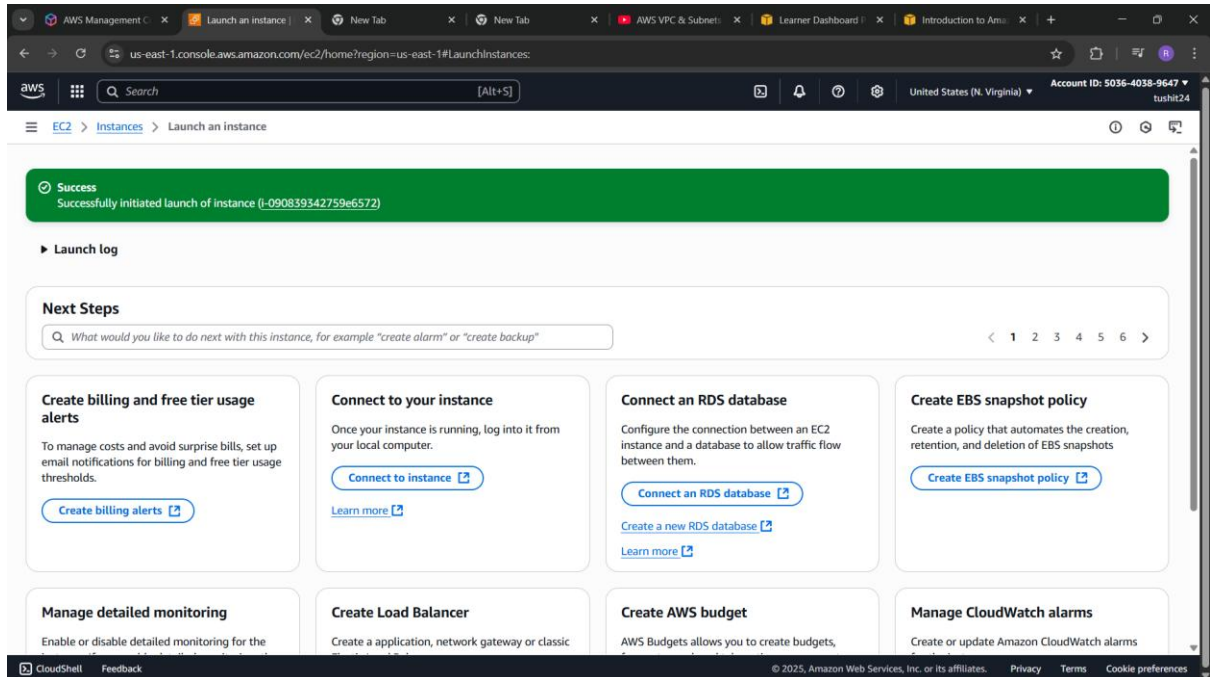
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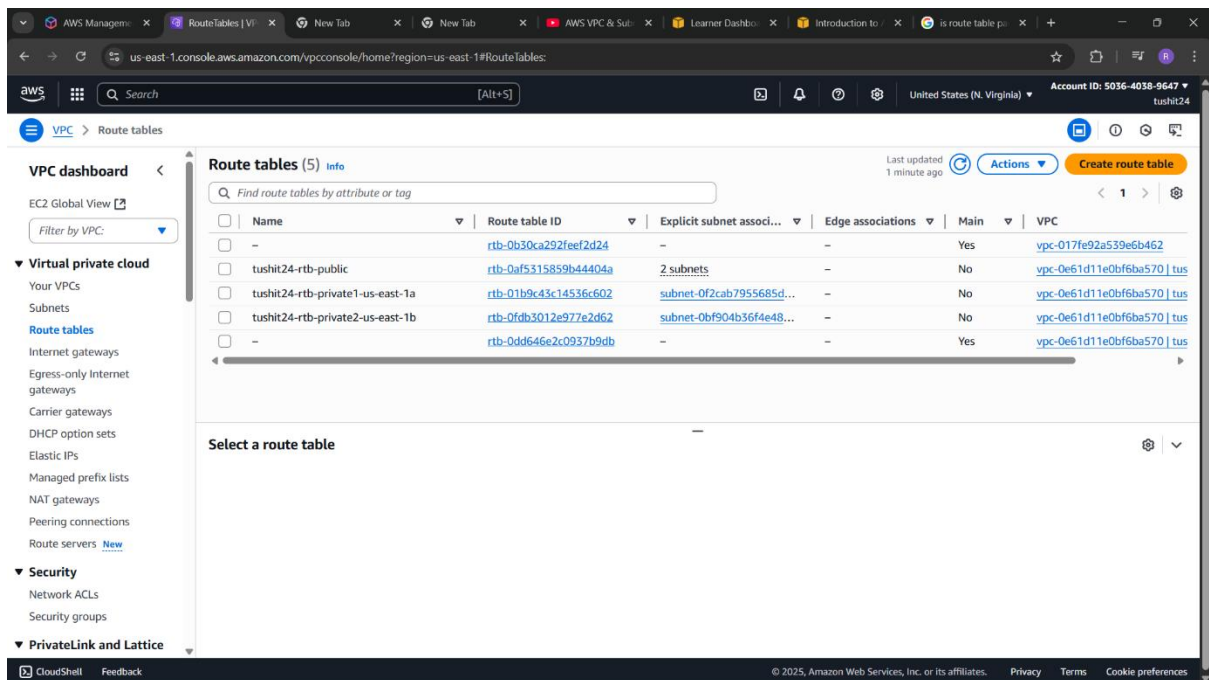
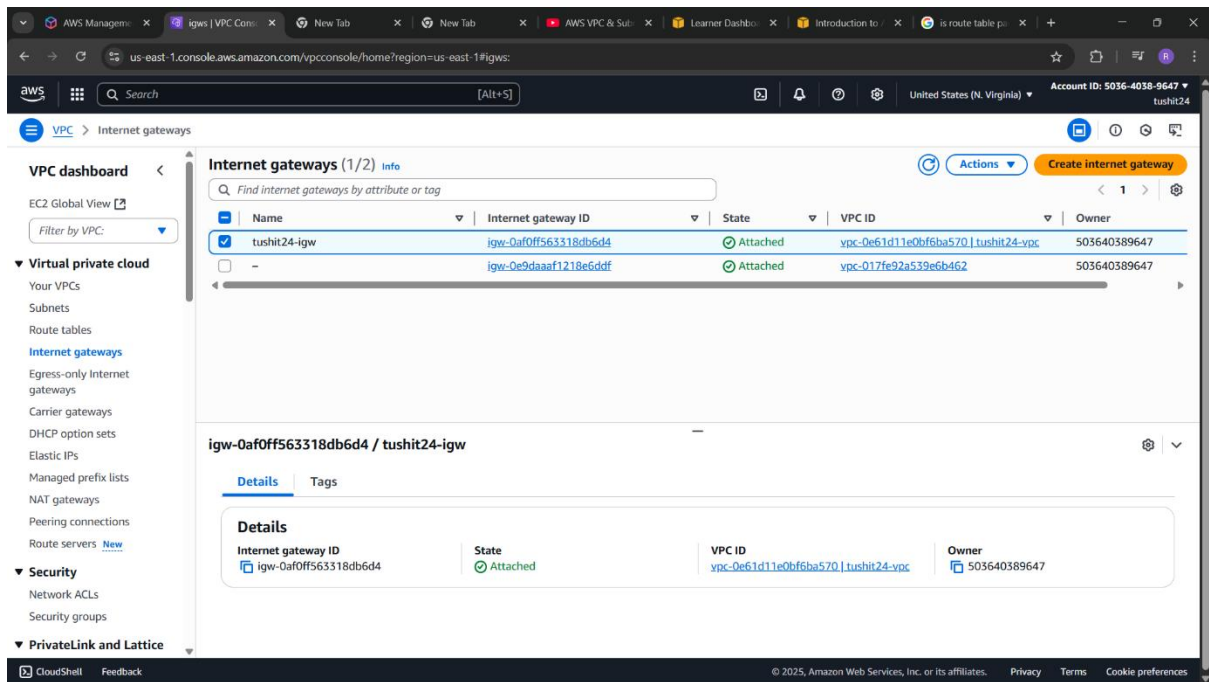
[Cancel](#) [Launch instance](#) [Preview code](#)

Configure & Launch EC2(private and public)

Steps:

1. Configure security group (allow SSH/HTTP).
2. Create or select a key pair.
3. Review instance configuration.
4. Click on **Launch Instance**.
5. Verify EC2 instance is running in the same VPC.





Internet Gateway and Route Table Configuration

Steps:

1. Go to **VPC Dashboard** and create an Internet Gateway.
2. Attach the Internet Gateway to your created VPC.
3. Navigate to **Route Tables** and create a new route table.
4. Add a route with destination **0.0.0.0/0** and select Internet Gateway as target.
5. Associate the route table with the subnet to enable internet access.

us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#RouteTables:

Search [Alt+S]

United States (N. Virginia) Account ID: 5036-4038-9647 tushit24

VPC dashboard

EC2 Global View

Filter by VPC:

- Virtual private cloud
 - Your VPCs
 - Subnets
 - Route tables
 - Internet gateways
 - Egress-only Internet gateways
 - Carrier gateways
 - DHCP option sets
 - Elastic IPs
 - Managed prefix lists
 - NAT gateways
 - Peering connections
 - Route servers
- Security
 - Network ACLs
 - Security groups
- PrivateLink and Lattice

Route tables (1/5)

Find route tables by attribute or tag

Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC
tushit24-rtb-public	rtb-0af5315859b44404a	2 subnets	-	No	vpc-0177fe92a539e6b462 tus
tushit24-rtb-private1-us-east-1a	rtb-01b9c43c14536c602	subnet-0f2cab7955685d...	-	No	vpc-0e61d11e0bf6ba570 tus
tushit24-rtb-private2-us-east-1b	rtb-0fdb3012e977e2d62	subnet-0bf904b36f4e48...	-	No	vpc-0e61d11e0bf6ba570 tus
-	rtb-0dd646e2c0937b9db	-	-	Yes	vpc-0e61d11e0bf6ba570 tus

rtb-0af5315859b44404a / tushit24-rtb-public

Details Routes Subnet associations Edge associations Route propagation Tags

Routes (2)

Filter routes

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-0af0ff563318db6d4	Active	No	Create Route
10.0.0.0/16	local	Active	No	Create Route Table

us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#vpcs:

Search [Alt+S]

United States (N. Virginia) Account ID: 5036-4038-9647 tushit24

VPC dashboard

EC2 Global View

Filter by VPC:

- Virtual private cloud
 - Your VPCs
 - Subnets
 - Route tables
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 - Carrier gateways
 - DHCP option sets
 - Elastic IPs
 - Managed prefix lists
 - NAT gateways
 - Peering connections
 - Route servers
- Security
 - Network ACLs
 - Security groups
- PrivateLink and Lattice

Your VPCs (1/2)

Find VPCs by attribute or tag

Name	VPC ID	State	Block Public...	IPv4
-	vpc-0177fe92a539e6b462	Available	Off	172.31.0.0/16
tushit24-vpc	vpc-0e61d11e0bf6ba570	Available	Off	10.0.0.0/16

vpc-0e61d11e0bf6ba570 / tushit24-vpc

Details Resource map CIDRs Flow logs Tags Integrations

Details

VPC ID vpc-0e61d11e0bf6ba570	State Available	Block Public Access Off	DNS hostnames Enabled
DNS resolution Enabled	Tenancy default	DHCP option set dopt-0c337b9416624924d	Main route table rtb-0dd646e2c0937b9db
Main network ACL acl-02a1a5c6e161955...	Default VPC vpc-09c59b09	IPv4 CIDR 10.0.0.0/16	IPv6 pool

Delete the EC2 instances and VPC

Terminate (delete) instances

On an EBS-backed instance, the default action is for the root EBS volume to be deleted when the instance is terminated. Storage on any local drives will be lost.

Are you sure you want to terminate these instances?

Instance ID	Termination protection
i-0119da3a18784fbf4 (tushit24-public)	Disabled
i-090839342759e6572 (tushit24-private)	Disabled

To confirm that you want to delete the instances, choose the terminate button below. Instances with termination protection enabled will not be terminated. Terminating the instance cannot be undone.

Skip OS shutdown
This option skips the graceful OS shutdown process. Use only when your instance must be stopped immediately, such as during an emergency or failover.
☐ Skip OS shutdown

[Cancel](#) [Terminate \(delete\)](#)

Successfully initiated termination (deletion) of i-0119da3a18784fbf4, i-090839342759e6572

Instances (2/3)

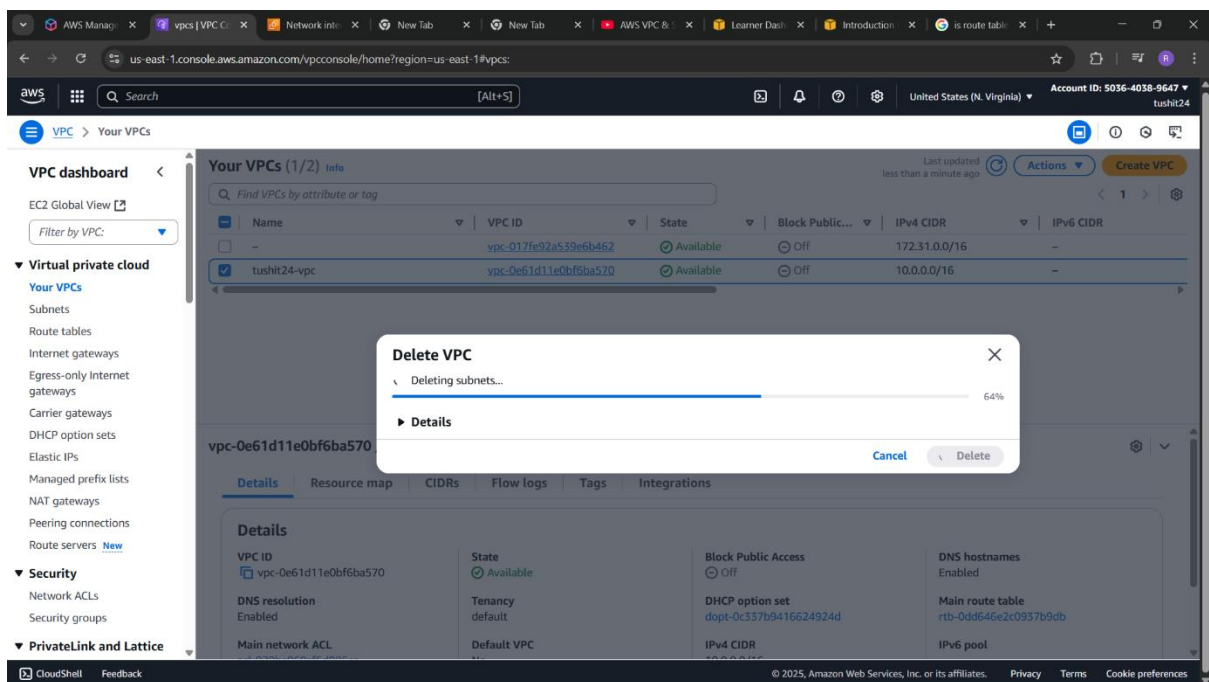
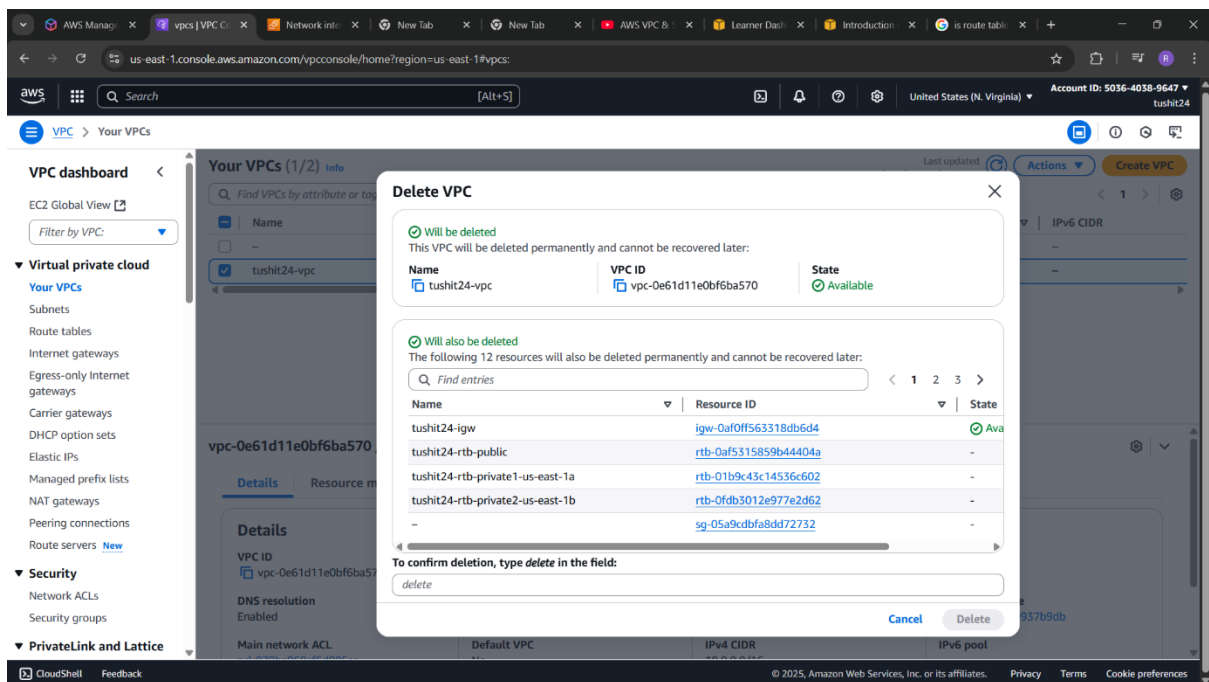
Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IP
tushit24-public	i-0119da3a18784fbf4	Shutting-down...	t2.micro	2/2 checks passed	View alarms +	us-east-1b	-
tushit24-private	i-090839342759e6572	Shutting-down...	t2.micro	2/2 checks passed	View alarms +	us-east-1a	-
tushit24	i-0d12dd971ca0e5448	Terminated	t2.micro	-	View alarms +	us-east-1a	-

2 instances selected

Monitoring

Alarm recommendations [Investigate with AI - new](#) 1h 3h 12h 1d 3d 1w Custom UTC timezone [Explore related](#)

CPU utilization (%) 1.25% 0.625%
Network in (bytes) 5.34k 2.67k
Network out (bytes) 4.99k 2.56k
Network packets in (count) 43.5 21.8



us-east-1.console.aws.amazon.com/vpconsole/home?region=us-east-1/vpcs

Search [Alt+S]

United States (N. Virginia) Account ID: 5036-4038-9647 tushit24

VPC > Your VPCs

VPC dashboard

EC2 Global View

Filter by VPC:

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only Internet gateways

Carrier gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

Security

Network ACLs

Security groups

PrivateLink and Lattice

You successfully deleted vpc-0e61d11e0bf6ba570 / tushit24-vpc and 12 other resources.

Details

Your VPCs (1)

Find VPCs by attribute or tag

Name	VPC ID	State	Block Public...	IPv4 CIDR	IPv6 CIDR
-	vpc-017fe92a539e6b462	Available	Off	172.31.0.0/16	-

vpc-0e61d11e0bf6ba570 / tushit24-vpc

Details

Resource map

CIDRs

Flow logs

Tags

Integrations

VPC ID

vpc-0e61d11e0bf6ba570

DNS resolution

Enabled

Main network ACL

vpc-0e61d11e0bf6ba570

State

Available

Tenancy

default

Default VPC

No

Block Public Access

Off

DHCP option set

dopt-0c337b9416624924d

IPv4 CIDR

172.31.0.0/16

DNS hostnames

Enabled

Main route table

rtb-0dd646e2c0937b9db

IPv6 pool

-

CloudShell

Feedback

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